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Goofy vs. Regular: Laterality effects in surfing

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ABSTRACT

The aim of the present study was to test if lateral preferences of surfers are associated with behaviour and performance depending on the direction of a breaking wave. We hypothesized that wave direction and surf stance interact in creating favourable or debilitating performance demands as surfers are either facing the wave (frontside) or the wave is breaking in the back of the surfers (backside). Study 1 was an online survey collecting self-report data of recreational surfers ($n = 394$). In Study 2, we analysed all wave scores ($n = 2,552$) and laterality of professional surfers during the season of 2014. Study 1 demonstrated that recreational surfers preferred surfing frontside and described themselves as more skilful when surfing frontside as this is facilitative for picking up visual information. Study 2 did not provide clear evidence that professional surfers on average scored higher during contests when surfing frontside, but when professional surfers had a choice of surfing frontside vs. backside, they were more likely to surf frontside. We discuss the diverging findings between Study 1 and Study 2 from the “circumvention-of-limits” argumentation within the expertise literature as professional surfers most likely have acquired skills allowing them to compensate for debilitating individual and environmental circumstances.

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Kurt Lewin (1936) is often quoted for stating that two different individuals placed in the exact same environment will not necessarily engage in the same behaviour. Or stated differently, even when the environment is identical for the physicist, the situation can be perceived as fundamentally different depending on the individual. Lewin formalized this idea in the formula $B = f(P, E)$ which states that behaviour (B) is a function of both the person (P) and the environment (E). Individual people behave entirely differently under the same environmental conditions depending on physical, psychological, and cultural characteristics of that person. In the present paper, we investigate

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how lateral preference as an instance of a personal variable interacts with environmental factors to shape human behaviour in the field of wave surfing.

Humans have lateral preferences as they predominately use either side of the body to carry out specific tasks. Although human handedness is the most prominent example of lateral preference (McManus, 2002), the preferred stance orientation when for example snowboarding, surfing, boxing, or fencing is another instance of lateral preference (Loffing & Hagemann, 2015; Loffing, Sölter, & Hagemann, 2014; Raymond, Pontier, Dufour, & Møller, 1996). While performance effects of stance orientation have been investigated in combat sports (e.g., Baker & Schorer, 2013; Loffing & Hagemann, 2015), they have not been investigated in surfing. Hence, we test whether lateral preferences in surfing can be beneficial for meeting environmental task demands.

Similar to snowboarding or skateboarding, surfers stand either with the left foot (regular stance) or with the right foot ("goofy" stance) in front when surfing waves on a surfboard (Warshaw, 2003). Pertinent to the present research, one characteristic of breaking waves is that they break either to the left- or to the right-hand side (when surfing towards the beach from the water) depending on a range of environmental factors (Butt, 2014). The direction to which a wave breaks determines the direction to which the surfer surfs the wave (cf. Figure 1). Right-breaking waves are surfed to the right and left-breaking waves to the left (perspective from the water when looking towards the beach).

The top left panel of Figure 1 displays a regular-footed surfer surfing a wave breaking to the right which is referred to as surfing frontside because the torso of the surfer is facing the unbroken section of the wave in surfing



Figure 1. Surfing stance and wave direction. The left panels represent right-breaking waves and the right panels left-breaking waves.

direction. When the back of the surfer is oriented towards the unbroken section of the wave, it is referred to as backside surfing (top panel right and bottom panel left of [Figure 1](#)).

The central hypothesis we test in the present research is if surfing stance and wave direction interact in creating facilitative conditions for surfing when surfing frontside as compared to backside. The reason for this hypothesis is that surfing frontside seems beneficial for picking up relevant visual information to guide the motor actions necessary for successful surfing. This is due to the fact that during frontside surfing, surfers are facing the unbroken section of the wave in the direction they are surfing while they have to turn their head and torso when surfing backside to pick up important visual information. There is a large body of evidence demonstrating the necessity of visual information in guiding skilful actions (e.g., Land & McLeod, 2000; Prinz, 1997). Neuroscience research has suggested that the neural mechanisms regulating goal-directed movements benefit from the accurate and timely spatial information of foveal information (Land, 2009). Consequently, performance will decline if accurate and timely spatial information is harder to acquire or missing. Of further relevance to this argument is that perception and action are believed to be tightly coupled in the regulation of skilled coordinated movements as this tight coupling is necessary to ensure stable action patterns that are appropriate to the specific environmental demands of the situation (Bertenthal, 1996; Farrow & Abernethy, 2003; Thelen, 1990). Arguably the perception-action-couplings in frontside surfing are facilitative for the motor response synergies necessary (Thelen, 1990) for skilful surfing which are modulated by perceptual information specifying the breaking wave and the spatial orientation of the surfer. The following quote supports this reasoning in surfing:

It is possible that different wave characteristics better fit particular surfers' technical skills. For example, "rights" (i.e., waves that break from the "peak" to the surfer's right) might be better for "regular footer" surfers [...] than for "goofy footer" surfers [...]. This is because regular footer surfers are gliding over the wave facing their look to the wall of the wave, which is believed to be advantageous for fast and precise surfing. (Mendez-Villanueva, Mujika, & Bishop, 2010, p. 138)

Hence, we hypothesized that surfers should prefer surfing frontside, chose this option more often when given a choice, and be able to perform better when facing the wall of the breaking wave as opposed to when the wave is breaking with the surfer's back towards the wall and having to look sideways and behind to see the wall (surfing backside). In order to test this hypothesis, we first conducted an online survey collecting self-report data on lateral preferences on non-professional (i.e., recreational) surfers' lateral preferences in surfing and second analysed wave scores and laterality of professional

surfers during the season of 2014 in the world title championships of the Association of Surfing Professionals (ASP; today known as WSL – World Surf League).

Study 1

Method

Participants

Three hundred and ninety-nine surfers completed an online survey. Out of the 399 data sets, five were excluded from further analysis due to being “active pro surfers,” leaving $n = 394$ for further analyses (23% female). The mean age was 36.03 ± 14.36 years and the mean surf experience was 16.11 ± 15.76 years. The sample of surfers was comprised from 29 different nations (97 from USA; 48 from Germany; 45 from Australia; 42 from the UK; 42 from New Zealand; 17 from South Africa; 13 from France; 12 from Belgium; 10 from Canada and the Netherlands; while the remaining nations had below 10 participants). Informed consent was obtained online. The study was approved by the Ethics Board of the German Sport University Cologne.

Procedure and measures

An online survey was distributed for a six-week period among recreational surfers through social media and surf clubs worldwide. The questionnaire collected general surf data as well as lateral preferences towards either a front-side or backside surf stance. To improve the reliability of the data, surf-specific terms were explained to the subjects in both writing and picture forms.¹ The experimental protocol was administered via SoSci Survey (Leiner, 2014).

Items assessing surfing profile, experience, and ability level. The first set of items assessed surfing experience (e.g., I started surfing XX years ago; On average, each year I went surfing on (a) less than 10 days; (b) 10–30 days; (c) 31–60 days; [...] (g) over 150 days) and ability level (e.g., after the take-off I have total control towards me and my board most of the time on a 7-point Likert scale with the poles completely disagree to completely agree).

Items assessing surfing stance. An image was included to illustrate the two different surf stances. Three items assessed the stance of the participants for surfing, skateboarding, and snowboarding with the following forced choice options: (a) goofy (right foot forward); (b) regular/natural (left foot forward); (c) don't know/never tried it.

Items assessing surfing preferences and abilities regarding stance. The first item in this section asked participants in a forced choice format

¹The full online questionnaire is available in the online supplementary material and we only describe the variables directly related to the research question in the manuscript.

whether they preferred to surf to one side with the answer options: (a) yes, frontside because ... ; (b) yes, backside because ... ; (c) depends on ... ; (d) no, because ... ; (e) don't know. After participants clicked on an answer, a text box appeared in which they were required to state the reason for their answer. The next three items were visual analogue scales with the poles frontside and backside with a default button placed to the right of the digital slider labelled equally. To give their ratings, participants had to click on the continuous scale and move the digital slider to either the frontside or backside pole. The software transferred the final position in a value between 0 (frontside) and 100 (backside). When the participants did not have a backside or frontside preference they clicked on the no preference button which was transferred to a value of 50. The three items were: (a) I wipe out more often while surfing ... ; (b) I have more control while surfing ... ; (c) I have more fun while surfing The next five items used the same digital slider format as described above and asked the following questions. I am generally better at (a) executing basic manoeuvres (i.e., bottom-turn; top-turn) while surfing ... ; (b) seeing the wave breaking while surfing ... ; (c) generating speed on the wave while surfing ... ; (d) executing advanced manoeuvres (i.e., floater; re-entry) while surfing ... ; (e) stalling for a tube-ride while surfing

Results and Discussion

Preliminary analyses

Surf stance was reported to be 58.2% regular-footed and 41.3% goofy-footed (0.5% unknown). This distribution was similar for skateboarding and snowboarding, although approximately 7% stated that they had not tried skateboarding and 19% had not tried snowboarding. No significant differences were evident in these distributions between male and female surfers. Experience and ability level did not significantly affect this distribution.

Main analyses

A Chi-square test revealed that "frontside" had a significantly ($\chi^2 (4, n = 373) = 242.08; p < .001; r = 0.81$) higher answer rate with 50.13% over the remaining possibilities (backside 6.7%; no preference 17.43%; depends on 20.11%; don't know 5.63%). As it might not be reasonable to assume an equal distribution for all five answer possibilities, a binominal test on the frontside and backside preferences demonstrated a significantly higher proportion ($p < .001$) of frontside compared to backside responses. The main reasons for a frontside preference were easier to execute manoeuvres ($n = 31$); better perception of the wave ($n = 31$); better control ($n = 30$); easier to surf ($n = 27$); feeling ($n = 26$); more natural ($n = 18$); easier to generate speed ($n = 15$); better skills ($n = 13$); others ($n = 22$).

A series of one sample *t*-tests (all two-tailed) were conducted to test whether participants' skill ratings differed from a value of 50 which indicated no lateral preference towards either frontside or backside. As shown in Figure 2(A), participants stated that they were generally better frontside at executing basic manoeuvres ($t(357) = -9.048, p < .0001, d = 0.478, 95\% \text{ CI } [0.369, 0.587]$), seeing the wave ($t(323) = -11.894, p < .0001, d = 0.661, 95\% \text{ CI } [0.54, 0.78]$), at generating speed ($t(315) = -13.693, p < .0001, d = 0.77, 95\% \text{ CI } [0.644, 0.895]$), executing advanced manoeuvres ($t(246) = -10.819, p < .0001, d = 0.688, 95\% \text{ CI } [0.549, 0.826]$), and tube riding ($t(238) = -6.676, p < .0001, d = 0.432, 95\% \text{ CI } [0.30, 0.564]$). Further, participants stated that they were less likely to wipe out surfing frontside ($t(279) = 5.979, p < .0001, d = 0.357, 95\% \text{ CI } [0.236, 0.478]$), in more control ($t(310) = -12.152, p < .0001, d = 0.689, 95\% \text{ CI } [0.565, 0.812]$); and that they have more fun ($t(252) = -7.202, p < .0001, d = 0.453, 95\% \text{ CI } [0.324, 0.581]$) (see Figure 2(B)).

The online survey supports the hypothesis that surfers prefer surfing frontside and that they are able to perform better (based on self-reports) when surfing frontside as opposed to backside. Surfing frontside is described as beneficial for their surfing and, more specifically, surfers reported to have better skills when surfing frontside as they are more in control of executing manoeuvres when their torso is oriented to the breaking wave which is facilitative to pick up important visual information in guiding and planning actions.

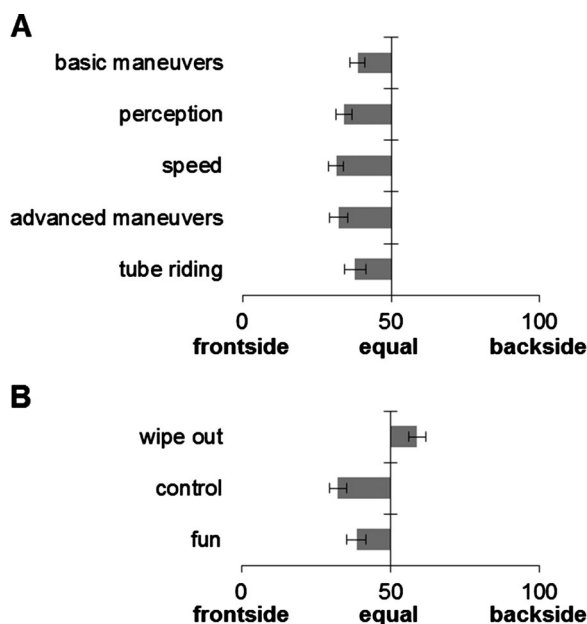


Figure 2. Mean ratings ($\pm 95\%$ confidence intervals) of surfer's described (A) abilities and (B) experiences depending on surfing backside or frontside (Study 1).

In view of these self-report findings, it seems reasonable to assume that lateral preferences in surfing would be associated with measurable variation in behaviour and performance in dependency of wave direction during surfing contests. This is because the preferred and more skilful option of surfing frontside depends on a surfer's lateral preference and wave direction (cf. Figure 1). Therefore, we analysed the season of 2014 in the world title championships of the Association of Surfing Professionals (ASP; today known as WSL – World Surf League).

Study 2

The wave scores for each ridden wave are recorded and published on the WSL website, therefore allowing analyses of wave scores as a function of wave direction and stance of the surfer surfing the wave. We hypothesized that (a) frontside surfing would on average result in higher wave scores and (b) surfers would more often chose to surf frontside in contests in which they had some degree of choice between frontside and backside waves. We deliberately state “some degree of choice” as during contests or heats it is often not the case that the right and left waves have a similar quality/scoring potential.

Method

Procedure and measures

We analysed wave scores of professional surfers during the season of 2014 in the world title championships of the ASP.² The ASP shares the relevant data online (<http://www.worldsurfleague.com/events/2014/mct>). The world title of professional surfing is decided throughout a season with multiple events taking place at different locations worldwide and consisted of 11 world tour events in 2014 (cf. Table 1).

The following variables were derived from the ASP homepage and the individual contest homepages.

Surfers' stance. Video material of the professional surfers was used to determine their stance by coding whether the right or left foot was at the back of the surfboard. During 2014, world title contests surfers never switched their surf stance.

Wave direction. Every heat of the season is video recorded and available online on the ASP (WSL) website and was used to determine whether the wave was breaking left or right.

Wave score. The individual wave scores were recorded from the contest homepages. Every wave an athlete surfed during a heat is given a score

²Not every heat has to finish with each surfer scoring two waves, e.g. due to injuries, heat rules or ill fate. Therefore, our total number of wave scores is not an exact product of event heats.

Table 1. Chronological order of event sites and spot information.

Surf spot	Location	Break type	Bottom	Direction	Winner stance
Snapper Rocks	Gold Coast, QLD, Australia	Point break	Sand/Rocks	right	goofy
Surfers' Point	Margaret River, Western Australia	Reef/Rocky	Reef	right	regular
Bells Beach	Torquay, Victoria, Australia	Reef/Rocky	Flat rocks	right	regular
Postinho	Rio de Janeiro, Brazil	Beach break	Sand	left and right	regular
Cloudbreak	Namotu, Tavarua, Fiji	Reef break	Reef	left	goofy
J-Bay	Kouga, Eastern Cape, South Africa	Point break	Sand/Rocks	right	regular
Teahupo'o	Tahiti, French Polynesia	Reef break	Reef	left	goofy
Trestles	San Clemente, California, USA	Point break	Sand/Rocks	left and right	regular
Le Gardian	Landes, Aquitaine, France	Beach break	Sand	left and right	regular
Supertubos	Peniche, Cascais, Portugal	Beach break	Sand	left and right	regular
Pipeline/Backdoor	Oahu, Hawaii, USA	Reef break	Reef	left and right	regular

between 0 points and 10 points. The scoring is done by a judging panel consisting of five judges, each declaring one score for a surfer's wave. The highest and lowest score gets eliminated and the mean of the remaining three scores determines the surfer's final wave score. The two best wave scores are summed at the end of the heat to comprise the final heat score. We only included these two highest heat scores of surfers in the present analyses. Altogether, 2,552 observations with information on both wave direction (1,679 right-breaking, 873 left-breaking) and wave score were available for analyses.

Surf direction. For the five events with waves breaking in both directions left and right (see Table 1) we analysed every wave surfed during a heat ($n = 4,218$ waves) instead of only the two top scoring waves and coded whether surfers went left vs. right.

Results and discussion

Preliminary analyses

Altogether 55 surfers (36 steadies, 19 *wildcard* entries) competed for the world title in the 2014 season, with 35 surfers being regular (63.6%) and 20 goofy (36.4%³). Of the 11 events, eight events were won by regular-footed surfers (five different surfers) and three by the same goofy-footed surfer (Gabriel Medina, later world champion).

³Of the 50 surfers competing on the qualifying tour for the professional tour in the 2014 season, 40 surfers were regular (80%) and 10 goofy (20%).

Main analysis

Two-factor univariate ANOVA on the scores of surfers with the between-subject factors stance (regular vs. goofy) and wave direction (left vs. right) revealed no significant main effect for stance, $F(1, 2,548) = 0.301$, $p = .583$, $\eta_p^2 < .001$, but a significant main effect for wave direction, $F(1, 2,548) = 9.253$, $p = .002$, $\eta_p^2 = .004$, as well as a significant interaction between stance and wave direction, $F(1, 2,548) = 4.297$, $p = .038$, $\eta_p^2 = .002$. Overall, both groups of surfers scored slightly higher on right-breaking than left-breaking waves. This pattern, however, was most pronounced in regular-footed surfers whereas goofy-footed surfers performed comparably on left- and right-breaking waves (see Figure 3(A)). Follow-up pairwise comparisons for the regular-footed surfers revealed a score difference in favour of right compared to left waves, $t(900.6) = 4.015$, $p < .001$, two-tailed; $d = 0.215$, whereas for goofy-footed surfers no difference was found, $t(600.4) = 0.561$, $p = .561$, two-tailed; $d = 0.04$.

Overall, the former analysis indicated an interaction between surf stance and wave direction; however, the effect size associated with the interaction (as well as for the main effect for wave direction) was very small. To further test the strength of evidence underlying the effects, we ran Bayesian ANOVA with the same factors as above along with a Bayesian independent samples t -test using the statistical software JASP (JASP Team, 2017). Accordingly, ANOVA findings supported the presence of the main effect for wave direction ($BF_{10} = 103.77$), but did not indicate evidence in favour of a two-way interaction ($BF_{10} = 0.563$). Nonetheless, regular-footed surfers' better performance in right- compared to left-breaking waves was supported, $BF_{10} = 589.3$, while for goofy-footed surfers evidence was in favour of no performance difference between left- and right-breaking waves, $BF_{10} = 0.095$.

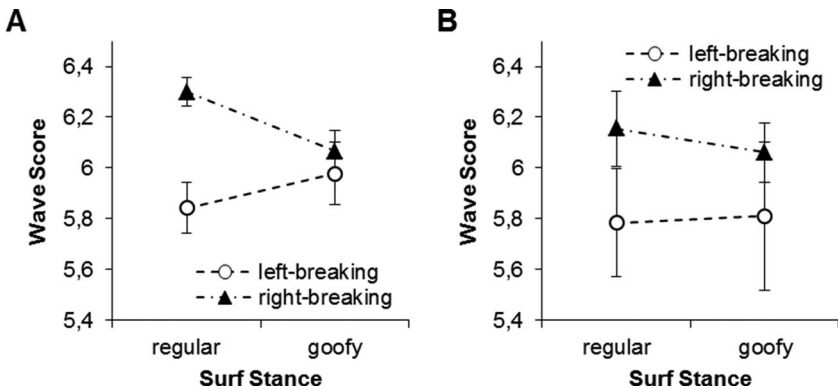


Figure 3. Mean wave score ($\pm$ SEM) as a function of surf stance and wave direction (A) across all observations ($N = 2,552$) and (B) only for surfers with at least 10 runs in left and right waves, respectively (Study 2).

The former results could be interpreted as limited, despite good statistical power (Schweizer & Furley, 2016), because all observations were treated as independent. However, the 55 different surfers were represented more than once in the dataset and therefore scores could be seen as dependent. To address this issue, we further used surfers instead of waves as unit of analysis and compared the average wave scores of every wave counting towards the total heat score in all heats of the 2014 season. We only included surfers who had competed in at least 10 left-breaking and 10 right-breaking waves in 2014 so that average scores for left- and right-breaking waves were a more reliable estimate of surfers' performances. Accordingly, 34 surfers (12 goofy) were included. As is illustrated in Figure 3(B), regular and goofy surfers again had a descriptively higher mean score for right- compared to left-breaking waves, $F(1, 32) = 4.076$, $p = .052$, $\eta_p^2 = .113$, $BF_{10} = 1.995$. However, there was no (trend of a) difference between surfing stance, $F(1, 32) = 0.018$, $p = .895$, $\eta_p^2 = .001$, $BF_{10} = 0.364$, and also no indication of an interaction between surf stance and wave direction, $F(1, 32) = 0.150$, $p = .702$, $\eta_p^2 = .005$, $BF_{10} = 0.371$. Thus, results show no clear evidence in favour of the hypothesized interaction between surf stance and wave direction. One possible reason why no clear evidence was found could be that our sample was composed of professional surfers who, as part of their expertise in surfing, might not be similarly affected by wave direction as might be the case in non-professionals who were interviewed in Study 1. This will be discussed further in the General Discussion.

There might be another potential indication that surf stance and wave direction are related, even in professional surfing. Namely, regular and goofy surfers might differ in the choice between right- and left-breaking waves when they have the opportunity to do so. That is, provided that professional surfers have the preference towards the (theoretically) better fit between surf stance and wave direction, then regular (goofy) surfers should more often decide to go right (left). We tested this theory by analysing if surf stance created a systematic bias in directional choices on the five contests that have left and right waves on offer. To this end, it was necessary to analyse every wave surfed in every heat ($n = 4,218$ waves) instead of just analysing the two top scoring waves that counted towards the heat total since performance and left- vs. right-choice might be confounded in the two top scoring waves. A Chi-squared test demonstrated an uneven distribution of left and right waves surfed depending on the stance of the surfer $\chi^2(1, N = 4,218) = 62.39$; $p < .001$, two-tailed, $r = .12$, $BF_{10-Poisson} = 1.688 \times 10^{12}$ (Jamil, Ly, Morey, Love, Marsman, & Wagenmakers, 2017). Specifically, goofy-footed surfers surfed left waves in 47.6% of the cases and right waves in 52.4%, whereas regular-footed surfers surfed left waves in only 34.7% of the cases and right waves in 65.3%. Based on the odds ratio of 1.71 (95% CI [1.50, 1.96]), goofy-footed surfers were about 1.7 times more likely than regular-footed surfers going left as opposed to right in contests that had left and right waves on offer.

General discussion

The aim of the present study was to test if lateral preferences of surfers are associated with behaviour and performance depending on the direction of breaking waves. Based on self-report data, Study 1 provided evidence that recreational surfers are influenced by the interaction of their personal surf stance and the environmental factor wave direction as they preferred surfing frontside in comparison to backside and described themselves as more skilful when surfing frontside. The pattern of results for the retrospective analyses of the 2014 season of professional surfing was not as straight forward concerning our hypothesis. While there was tentative evidence that professional surfers on average scored higher during contests when surfing frontside in comparison to backside, this pattern did not hold up in further analyses. However, when professional surfers had a choice of surfing frontside vs. backside during particular contests, they were more likely to surf frontside and this was particularly evident in regular-footed surfers.

Taken together, the present research shows that stance orientation can affect surfing and potentially can have performance advantages depending on the direction the wave is breaking in. However, at the highest level of professional surfing during the 2014 season we did not find strong evidence for a performance advantage for frontside surfing in comparison to backside surfing. One reason for this is that professional surfers most likely have acquired skills and strategies to circumvent potential performance limitations due to personal characteristics like physical size, body composition or surf stance. This process has been termed “circumvention-of-limits” within the expertise literature and has received support in a vast amount of empirical research in the cognitive domain (see e.g., Ericsson, Charness, Feltovich, & Hoffman, 2006; Hambrick, Macnamara, Campitelli, Ullén, & Mosing, 2016, for reviews). The basic argument is that athletes acquire compensatory skills and strategies through practice that increase information-processing efficiency. In turn, they are better able to meet the demands of challenging performance environments (e.g., Furley & Dörr, 2016, for the sport of surfing). According to Ericsson, Krampe, and Tesch-Römer (1993), these skills are (mainly) acquired through deliberate practice—“activities that have been specially designed to improve the current level of performance” (p. 368)—and serve to circumvent performance limitations associated with individual characteristics and basic abilities. In this respect, professional surfers most likely have acquired skills through deliberate practice activities that help them prevent performance decrements when surfing backside which can be considered an instance in which individual factors (stance) interact with environmental factors (wave direction) in creating more difficult task demands as the sampling of important visual information is harder due to the wave breaking in the back of the surfer (cf. Study 1).

The most frequent line of reasoning when discussing performance advantages associated with lateral preferences has been termed *fighting hypothesis* (Raymond et al., 1996), which argues for a fitness benefit for a certain lateral preference in one-on-one interactive contexts by showing that left-oriented athletes are underrepresented in the overall population and it is therefore less common to be confronted with this orientation in a fight. Hence, an opponent fighter will have less experience in this situation and will be less likely at anticipating for example punches and defending accordingly (Loffing & Hagemann, 2016; but see Baker & Schorer, 2013; Loffing & Hagemann, 2015). The present research suggests that a certain lateral preference can be beneficial (or debilitating) to meet environmental task constraints in non-interactive situations (see Eastwood, 1972, for a similar argument considering lateral preference in golfing and golf course characteristics)—at least for recreational surfers (Study 1). Importantly, Study 2 indicates that increasing skill level can help to compensate for difficult performance demands as there was no clear evidence that professional surfers performed substantially worse when surfing backside compared to frontside. This finding is in line with results of Carey et al. (2001) on footedness in professional soccer. In retrospective analyses of the soccer World Cup in France in 1998, both left and right footed professional players showed distinct side preferences for one foot, although they were on average equally skilled with their non-preferred foot as they were with their preferred foot (on the rare occasions when they used it). Hence, although athletes show pronounced lateral preferences, expert athletes can be similarly skilled with their non-preferred side of the body. Future research might want to investigate into the question of how lateral preference interacts with practice activities in sports and thereby influences skill acquisition (Whiteside, Buszard, Giblin, & Reid, 2016).

In conclusion, the present paper reports that lateral preference as an instance of a personal variable and wave direction as an instance of an environmental variable interact in shaping behaviour in surfing. Although the interaction of personal and situational variables is very complex in predicting surfing performance (see Farley, Abbiss, & Sheppard, 2017, for a review), the present paper highlights that lateral preference and wave direction are important factors to consider in this endeavour.

Disclosure statement

No potential conflict of interest was reported by the authors.

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